

Question: Determine the most suitable bearing state type from...

Ground Truth: Undesired Input



...

Token Energy-Based Uncertainty Quantification

... description \” Moder ate Inner ...

$$(\dots \text{NLL}_{\delta-2} + \text{NLL}_{\delta-1} + \text{NLL}_{\delta} + \text{NLL}_{\delta+1} + \text{NLL}_{\delta+2} \dots) / N_t$$

Uncertainty score falls within the predefined margin



Acceptable uncertainty, return generation



Token Energy-Based Uncertainty Quantification

... description \” Moder ate Inner ...

$$\left(\begin{matrix} \dots \text{NLL}_{\delta-2} & \text{NLL}_{\delta-1} & \text{NLL}_{\delta} & \text{NLL}_{\delta+1} & \text{NLL}_{\delta+2} \dots \\ \times + & \times + & \times + & \times + & + \times \end{matrix} \right) / N_t$$
$$\left[1 + (\omega r_{pb} - 1) 1_{\{k=\delta-1\}} \right]$$

Uncertainty score is beyond the predefined margin



Unacceptable uncertainty, refuse to answer

